

Solution Requirements

Date	29 September 2026
Team ID	SWTID-2026-7446
Project Name	ComicCrafts – AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High / Medium / Low)
1	Authentication	No login is required. ComicCrafts is a local web application, so users open the web page and start creating comics directly. The Gemini API key is stored securely on the server side.	Low
2	Authorization levels	Single user role (Comic Creator). The user can enter a story prompt, generate, preview and download comics. There is no admin role in the current version.	Low
3	External interfaces	Google Gemini API (Gemini 1.5 Pro / Flash) for the comic outline, story and dialogues. Stable Diffusion through Hugging Face Diffusers and PyTorch for panel images. FPDF library for PDF creation.	High
4	Transactions processing	The user submits the form (story prompt, character name, setting, tone, art style). FastAPI routes the request, Gemini generates the outline and story, Stable Diffusion generates images, and the result is sent to the preview page. No payments are involved.	High

S.No	Requirement Category	Requirement Description	Priority (High / Medium / Low)
5	Reporting	On-screen comic preview with panel titles, images and narration. Full comic export as a PDF (images + narrative text) saved with a timestamped filename, followed by an export success page.	High
6	Business rules, etc.	Workflow: user input, outline, story, image prompts, images, layout, preview, PDF export. Changing tone or art style regenerates the entire pipeline. Each panel must contain a title, an image and text.	Medium
7	Compliance to laws or regulations	No personal data or user accounts are stored. Usage of the Gemini API and Stable Diffusion model must follow their terms and licenses. API keys are kept out of the source code and frontend.	Medium
8	Other	Both a Jinja2/HTML web interface and FastAPI API endpoints are provided. Users can regenerate a comic with different preferences until satisfied.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Comic outline and story are generated quickly by Gemini. Image generation time depends on the local hardware (GPU or CPU). PDF export must be fast.	Outline + story in under 15 seconds; PDF export in under 5 seconds
2	Scalability	The application runs locally for single-user use, and the modular FastAPI code can be moved to a cloud server with more workers later.	Handles multiple sequential requests without crashing
3	Security & Data Privacy	API keys are stored in environment variables and are never exposed in the frontend. User inputs are validated before processing. No personal data is stored.	0 API keys in client-side code; all form inputs validated
4	Reliability & Availability	The system shows a clear error message if Gemini or Stable Diffusion fails, instead of crashing. The server keeps running while the app is in use.	Failed requests show a friendly error message; app stays available

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
5	Usability & Accessibility	Clean, simple HTML form with clear labels for prompt, character, setting, tone and art style. Non-artists and casual users can create a comic without training.	Comic generated in 3 steps: fill form, submit, view preview
6	Other	Maintainability: modular code (routes.py, layout_builder.py, exporters.py). Portability: runs on Windows and Linux with Python, pip and virtualenv.	Any module can be updated without changing the others